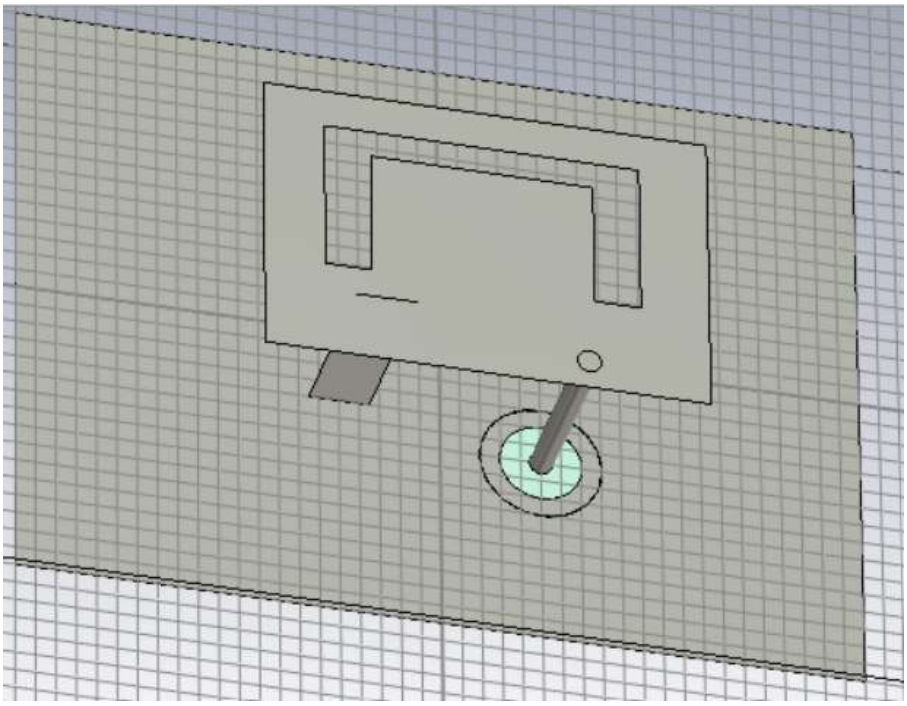
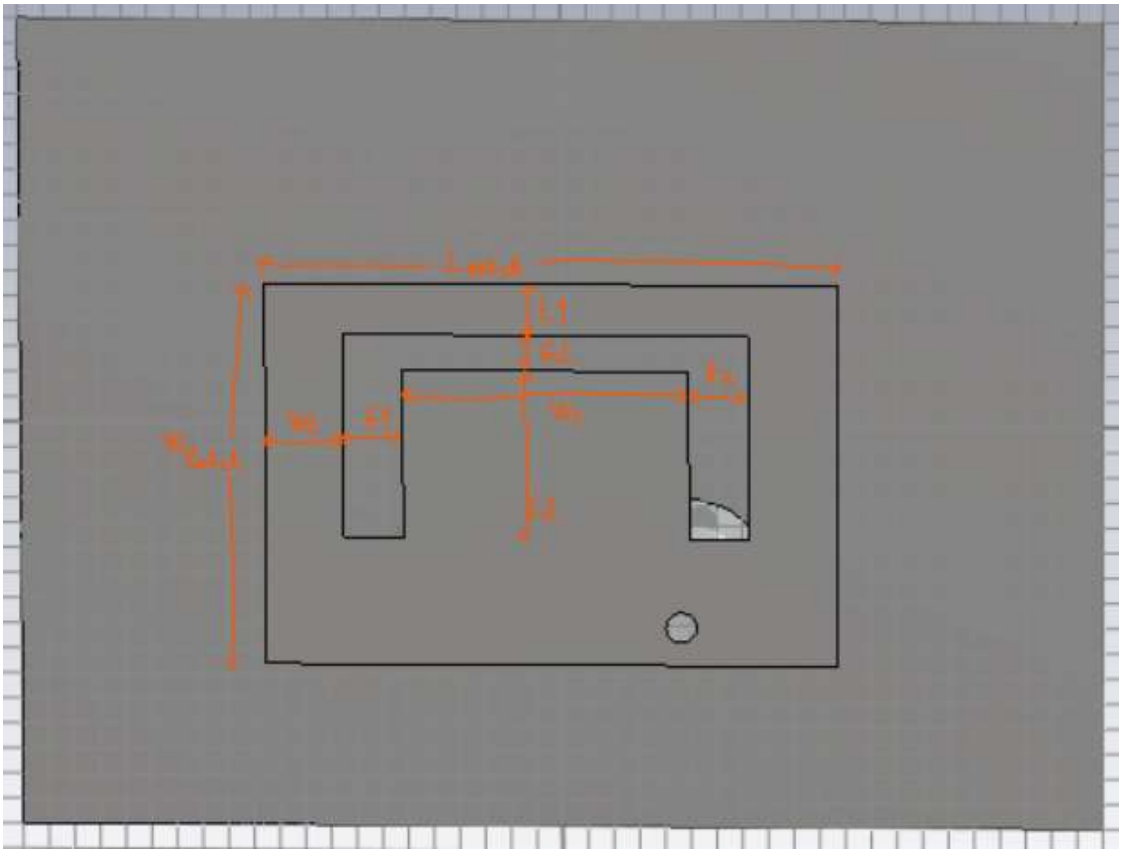
Sergi Salvà Pibernat
Eugeni Valentí Soler

Laboratory 4 Report WLA

Exercise 1:

Following the laboratory paper, we obtained the following schemmatic:



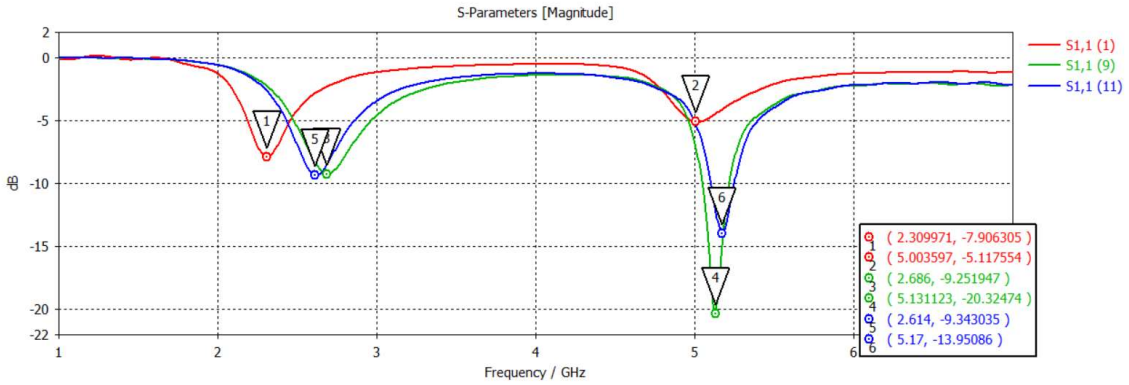


The following are the values introduced in the design in order to obtain the asked frequencies:

Parameter List

Name	Expression
coax_in	= 1.2
coax_out	= 4.1
G1	= 2.3
G2	= 1.4
H_antena	= 9
L1	= 2
L2	= 6.6
L3	= L_patch-L1-G2-L2
l_coax	= 10
L_ground	= 32
L_patch	= 15
l_strip	= 0
outer_width	= 1
W1	= 11
W2	= 3
W_ground	= 42
W_patch	= 22
W_strip	= 3
x0	= -5
x_strip	= -5
y0	= 6
y_strip	= -4

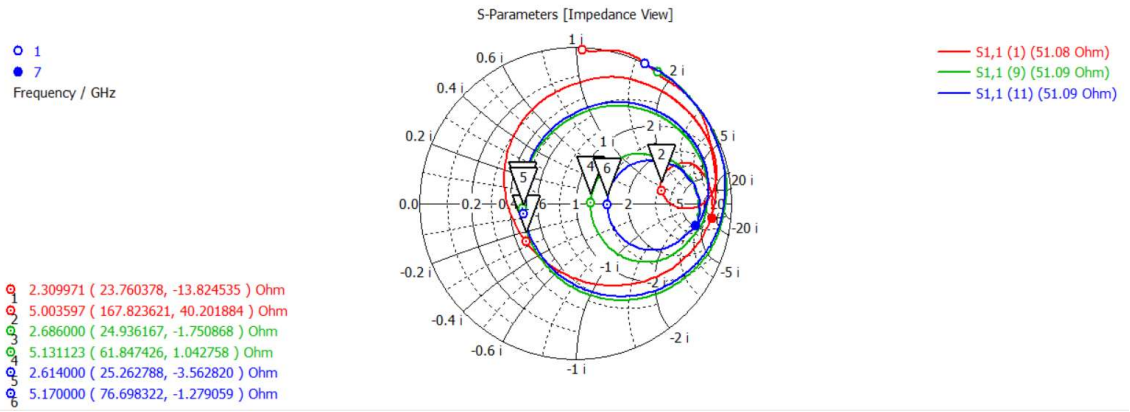
In the following graph we can observe the difference between the first try with the last try:



The graph has been obtained with the following values:

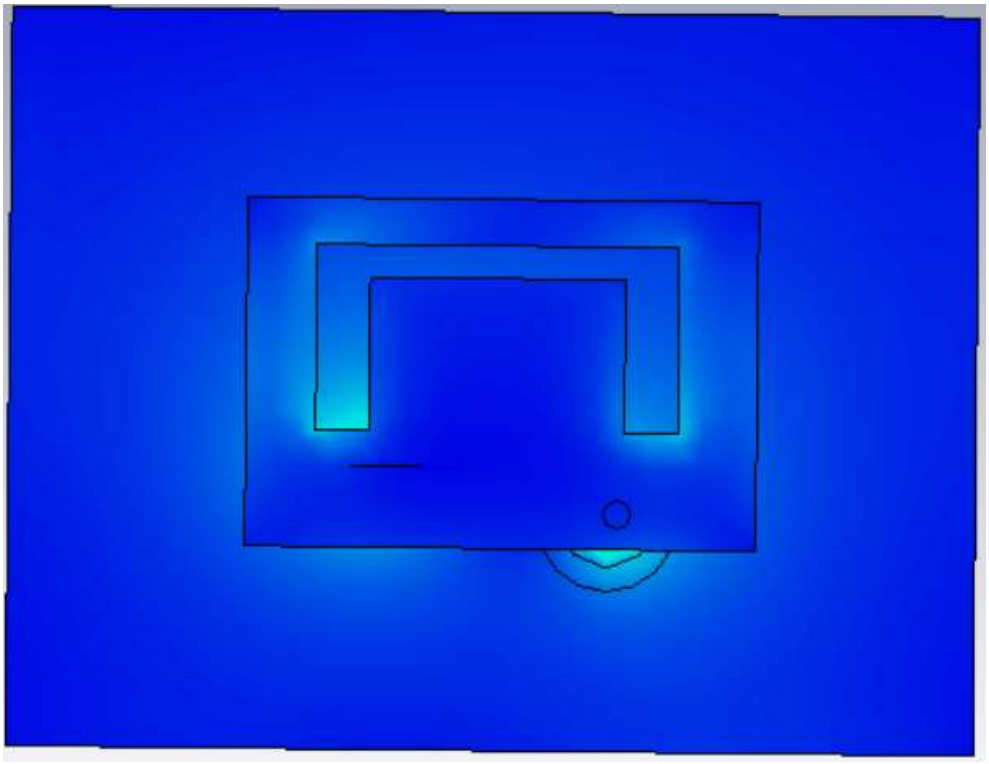
3D Run ID	x0	W1	W2	x_strip	y0	y_strip
1	2	11	3	-7	5	-4
9	-5	11	3	-7	6	-4
11	-5	11	3	-5	6	-4

We can observe that we have obtained a better marked frequency with a smaller bandwidth at the run 9.

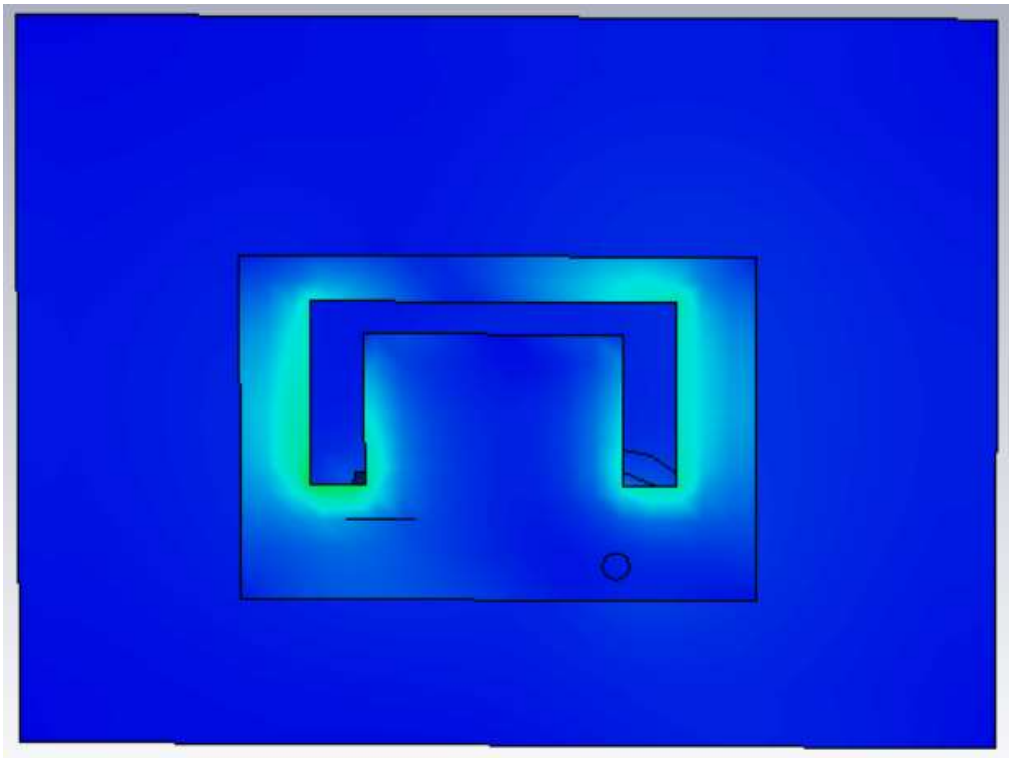


With the smitch chart we can see that we have improved the impedance approaching closer to the desired value of 50 Ohms.

In the following graphs we can observe the average H field for f=2.45GHz:



In the following graphs we can observe the average H field for f=5GHz:



With the animation we can observe that the patch oscillates at both frequencies.